

Polynomial Regression — Notes

1. What is Polynomial Regression?

Polynomial Regression is used when the relationship between **input (X) and output (y)** is **curved/non-linear** rather than a straight line.

Normal Linear Regression:

Polynomial Regression:

2. Main Idea

Instead of changing Linear Regression itself, we **create new features** from the existing feature.

For example:

Original:

X

Polynomial features:

X

X^2

X^3

Then Linear Regression uses these new features:

X, X^2 , X^3

↓

Linear Regression

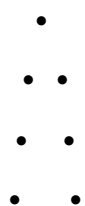
↓

Curved prediction

So even though the final model is technically **linear in its parameters**, it can produce a **non-linear curve** with respect to X.

3. Example

Suppose the actual relationship looks like:



A straight line won't fit it well.

Polynomial Regression can create a curve that follows the pattern much better.

4. PolynomialFeatures

In Scikit-Learn:

```
from sklearn.preprocessing import PolynomialFeatures
```

```
poly = PolynomialFeatures(degree=2)
```

```
X_poly = poly.fit_transform(X)
```

If:

```
X = [2, 3, 4]
```

degree = 2 creates features based on:

1, X , X^2

So:

2 → 1, 2, 4

3 → 1, 3, 9

4 → 1, 4, 16

Then:

```
model.fit(X_poly, y)
```

5. Choosing Degree

The **degree** controls how complex the curve can be.

Degree 1 → straight line

Degree 2 → simple curve

Degree 3 → more flexible curve

Degree 10 → very complex curve

Underfitting

Degree is too low → model is too simple → misses important patterns.

Overfitting

Degree is too high → model becomes too complex → learns **noise** instead of the real pattern.